

Payout policy

Week 43 — the manager's stake, and why the dividend waits for it

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Where we are

Wk	Friction	Effect	Evidence
36	Personal taxes (Miller)	value $\uparrow$ / $\downarrow$	net benefit (Korteweg, 2010)
37	Dynamics / flexibility	leverage $\downarrow$	flexibility survey (Graham and Harvey, 2001)
38	Costly external finance	investment $\downarrow$	cash hoarding (Almeida et al., 2004)
39	The manager wedge	risk-taking $\downarrow$	weak incentives (Jensen and Murphy, 1990)
41	Free-riding on control	takeovers $\downarrow$	acquirer gains ≈ 0 (Moeller et al., 2005)
42	Someone else's downside	hedging $\downarrow$ / risk $\uparrow$	gold miners (Tufano, 1996)
43	Unpledgeable cash flow	payout delayed	?

Week 39 asked what it costs to make a manager act. This week asks the same question **over and over**, and the answer changes — because a manager you deal with repeatedly can be made to build something he does not want to lose.

Today

- One question: what is the manager's stake in this firm, and where did it come from?
- He did not buy it. He earned it — and it turns out to be four things at once: the incentive device, the firm's financial slack, the thing whose ceiling triggers a dividend, and what the balance sheet you can observe is writing down.
- Two payoffs, and neither is assumed. **Payout** stops being a rule and becomes a boundary. And the **q-theory of investment**, which you met in Week 38, turns out to be misspecified in a way we can now name.

This is the last lecture, so it also closes the ledger — and it discharges an assumption Week 38 borrowed and never paid back.

What you own, and where it stops

Incentives, bought

- Week 39: the contract was $w = \alpha + \beta y$. The board **buys** incentives by handing over a share β of the outcome, and pays for it in a risk premium.
- Everything there happened **once**. One period, one contract, one draw of the outcome.
- You also own the payout benchmark: Miller and Modigliani (1961) — the split of earnings between dividends and retained earnings is irrelevant, and homemade dividends undo anything the firm does.

Neither is re-taught. But notice what they have in common: in both, **nothing carries over**. Today everything turns on what does.

Say the benchmark's prediction out loud

- Under Miller and Modigliani (1961) payout is the **residual** — pay out what is left. So the dividend should track **current earnings**, and a firm's history should not matter.
- And the manager's stake in the firm is **whatever you paid him this period**. Nothing accumulates, so nothing can be taken away.
- Both are wrong, and the same object fixes both.

Where this hour is going. Payout will turn out to depend on **how much the firm has accumulated**, not on what it earned this year. Watch for the moment the state variable arrives — everything after it is a consequence.

The threat of tomorrow

Financing is sequential, and that changes the problem

- Bolton and Scharfstein (1990): the manager sees the cash flow and can **divert** it. So a contract cannot simply demand the cash — there is nothing to enforce against.
- What can be enforced is the **future**. Refinance the firm next period **only if** this period's performance was good.
- **The threat of tomorrow disciplines today**. He hands over cash he could have taken, because taking it ends the relationship.

You have seen this without the name: staged venture rounds, covenants that trip on a ratio, a credit line the bank can decline to renew. The instrument is always the same — **continuation**, made conditional.

But it is a cliff

- One shot, and all-or-nothing: refinanced, or dead. There is no **position** in between and nothing carries forward.
- So the manager's stake is still a *flow* — what he gets this period — rather than a **stock** he has built and could lose.
- Which raises the obvious question, and it is the whole of the rest of the hour: **what if the relationship simply repeats?**

Bolton and Scharfstein are already almost dynamic — they have a tomorrow. What they do not yet have is a **yesterday**.

Repeat it, and the threat becomes a stock

The state variable

- Now the relationship runs on. Write W_t for the manager's **claim on everything the firm will pay him from here on** — his stake, carried forward.
- **It must move with the cash flow.** If a good year did not raise W and a bad year did not lower it, he would divert instead of reporting. The whole contract rests on that one requirement.
- λ is the fraction he could steal — **Week 38's pledgeability parameter seen from the other side**. There it said how much could be promised to outsiders; here it says how much can be taken.

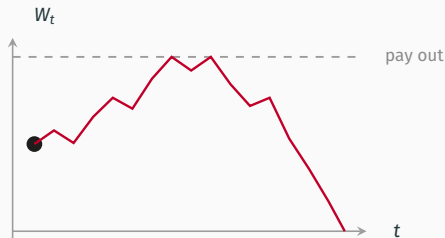
So his stake is not a number the board chose. It is a **running score**, and he is the one who moved it.

Two lines, and everything happens between them



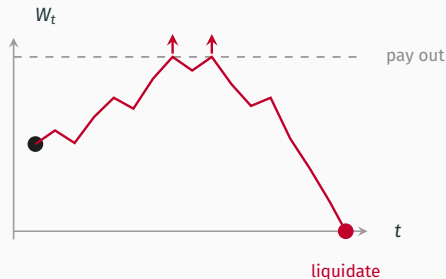
Vertical: the manager's stake W_t . It has a **floor** at zero — nothing left to lose, so nothing left to enforce with, and the firm is wound up. And it has a **ceiling**, for a reason we come to shortly. The dot is where a young firm starts.

His stake is the running score



Good years push it up, bad years pull it down — **it has to**, or he would divert. Note what this is not: it is not a payment schedule and not a performance rating. It is a **position**, and it is the only thing standing between him and liquidation.

Two kinds of event, and one firm has both



Twice the score reaches the ceiling and **cash is paid out** (the arrows). Years later it reaches the floor and the firm is **liquidated**. Same firm. A company that paid dividends and later died is not a puzzle here — it is one path.

What he is standing on

The manager's stake is **collateral he never had to post**. He did not bring it, and no board voted it to him — he **accumulated** it by performing, and he forfeits it if things go badly enough.

- So the distance from the floor is doing two jobs at once: it is **his exposure**, and it is **the firm's financial slack**. They are the same number.
- And it is self-tightening. A firm that has done well has a manager with a lot to lose, so he needs **less** watching — exactly when watching is cheapest.

Compare Week 39, and the contrast is the point of the lecture: there the board **bought** incentives and paid a risk premium for them. Here nobody bought anything.

Incentives are not bought. They are accumulated.

In Week 39 the only instrument was β — give him more of the outcome and he cares more, but he is risk-averse, so it costs. There was no way out of the trade-off, only a best point on it.

Here the instrument is **time**. Every good year moves him further from the boundary, and every step away relaxes the constraint that was binding. **The relationship itself does the work the pay package could not.**

Which is why a young firm and a mature one are not the same firm with different luck. They are at **different points of the same contract**, and almost everything we observe about them differs for that reason alone.

So payout is the ceiling

Why there is a ceiling at all

- Why not let his stake grow without limit? Because promising him more than the incentive problem **requires** is pure cost — **assume** he is less patient than the investors, and deferring his money destroys value. *That impatience is an assumption, and it is the one doing the work on this slide.*
- So there is a point $\bar{W}$ beyond which the firm stops accumulating and **pays cash out** instead.
- Below it: retain everything. At it: distribute. That is the entire payout policy, and **nobody wrote it down as a rule.**

Payout is a boundary, not a rule. It is what happens when the manager's accumulated stake hits its ceiling — so it is triggered by the **stock**, never by this year's earnings.

Three things that now follow without being assumed

- **Payout tracks accumulated slack, not current earnings.** Two firms with identical earnings this year pay differently if they arrived with different histories.
- **It is path-dependent.** Lintner (1956) interviewed managers who described exactly that and could not say why. Here it is not a habit — the state variable *is* the path.
- **Young firms do not pay; mature firms do.** Not a separate life-cycle theory. The same contract, read at two points.

Notice what never went in. Nothing so far has assumed that dividends are sticky, or costly to cut, or smoothed — and stickiness came out anyway. Assume the conclusion and you have a description wearing a derivation's clothes. But *something* went in. **Next slide.**

What we assumed, and what we did not

Assumed — three:

- cash flow is **divertible** at rate λ ;
- the manager is **less patient** than the investors;
- the refinance-if-good contract, **taken as given** from Bolton and Scharfstein (1990).

Derived:

- that his stake must move with the cash flow;
- that there is a **ceiling**, and payout happens at it;
- that payout is path-dependent, and young and mature firms differ;
- that investment rises with slack, and marginal q falls below average q .

Three assumptions is not zero and I am not claiming it is. The claim is narrower: **none of the three is a statement about payout**. Nothing here assumes dividends are sticky, costly to cut, or smoothed — and all three come out anyway.

Where W lives in a real balance sheet

- The contract can be implemented with securities you already know. In DeMarzo and Sannikov (2006) the manager's stake maps to the **unused capacity on a credit line** — draw it down on bad years, pay it back on good ones, and distribute only when it is fully repaid.
- Biais et al. (2007) implement the same contract with a **cash balance** instead.

Do not over-read this. The implementation is **not unique** — credit line and cash balance are two ways of writing the same object, and Sannikov (2013) says so explicitly. The robust statement is only that W corresponds to **some measure of financial slack**. That is enough for everything that follows.

And it breaks the q -regression

The other benchmark, said out loud

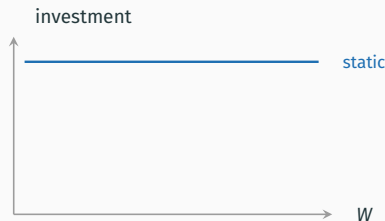
- **Hayashi (1982), the workhorse of investment.** Investment depends on Tobin's q and **nothing else**; and **average q equals marginal q** , so the ratio you can measure is the object the theory wants.
- That is not a caricature — it is the benchmark this literature actually runs on, and it is what every q -regression assumes.
- Set $\lambda = 0$ in our model — no diversion, no agency — and it **becomes** Hayashi. The static case is not a rival. It is nested.

So we can ask a sharp question: **what does turning λ back on do to a regression that assumes it is zero?**

- Fazzari et al. (1988) found investment responds to **cash flow** even controlling for q , and read it as evidence of financing constraints.
- Kaplan and Zingales (1997) objected that the sensitivity is not monotone in how constrained a firm is — and Week 38 re-derived that objection as a **prediction** of the collateral model.
- We left it there. Both sides were arguing about what a coefficient meant **inside a regression neither had derived**.

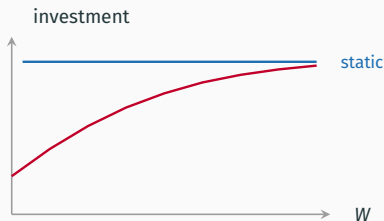
Now we can derive it.

What the static theory says investment depends on



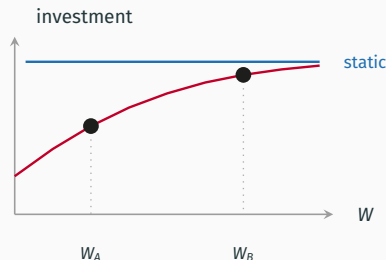
Horizontal: the firm's financial slack. Vertical: how much it invests. Under Hayashi the line is **flat** — for a given q , the balance sheet is irrelevant. Two firms facing the same opportunity invest the same amount, and that is the assumption the regression is built on.

What it actually depends on



The red curve is the same firm under the contract. Investment **rises with slack** and sits below the benchmark everywhere. The intuition is one sentence: a firm near the floor invests less **not because its projects got worse**, but because spending the buffer raises the chance of dying.

Same opportunity. Different investment.



Two firms, **identical q** , different slack — and they invest differently. DeMarzo et al. (2012) put it exactly so: investment moves with profits, past investment and slack **even with time-invariant investment opportunities**. Nothing about the opportunity changed.

What that does to the regression

Under dynamic agency, marginal q is below average q , and the wedge between them moves with financial slack. So average q — the thing you can measure — is a proxy whose measurement error varies across firms and over time.

- Put cash flow into that regression and of course it enters: it is correlated with the state variable the specification left out.
- The q -regression was never missing a control. It was missing a state variable. Fazzari and Kaplan–Zingales were both reading a coefficient that was carrying something neither had modelled.

And Week 38 pays its debt

- Week 38 wrote investment as $I = \min\{I^*, (1 + \lambda)w\}$ — a kink, from a collateral constraint that was **assumed**, with its microfoundation explicitly handed to the theory course.
- Today's curve is the same picture: investment rising in net worth, then flattening. But it came out of a **contract** — Week 38's net worth w is this week's W , and its pledgeability λ is this week's **diversion rate**.
- Same shape. One **borrowed**, one **earned**.

That is one loose end closed, and it is worth noticing what closed it: not new data, but taking the same friction and letting it run for more than one period. There is a second, and it belongs to Week 39.

And the same thing happens to pay

- **Static (Holmström 1979).** Filter out everything he did not control — strip the oil price out of an oil executive's bonus. Paying for luck is paying for **noise**.
- **The data disagree.** Bertrand and Mullainathan (2001): pay responds to oil prices, exchange rates and sector shocks — and **more where governance is weak**.
- **Hoffmann and Pfeil (2010):** if the shock is **persistent**, so today's luck predicts tomorrow's profitability, rewarding it can be **optimal**.

And note *where* it shows up. The incentive **slope** still filters luck out, exactly as Holmström says — what moves is **the level of his stake**. The same object again: luck that predicts future profitability should raise W . *Which does not make every observed reward for luck optimal — it removes the presumption that it must be capture.*

Evidence, and the close

The cleanest test is a variable nobody designed for this

- DeAngelo et al. (2006): the strongest predictor of whether a firm pays a dividend is the **earned/contributed capital mix** — retained earnings over total equity.
- Firms whose equity is mostly **earned** pay. Firms whose equity is mostly **contributed** pay almost nothing. It survives controls for profitability, growth, size, cash and dividend history, and holds separately for initiations and omissions.
- Read it against the model: retained earnings over total equity is **a proxy for how much the firm has accumulated**. It is W , measured with an accountant's ruler.

The life-cycle theory of dividends and the dynamic contract are not two explanations. One is the other, written in accounting data.

Read it like a referee

- Retained earnings over total equity proxies accumulated retention — but it also proxies **age**, **size** and **past profitability**, and those have their own stories.
- The model predicts the **whole path**, not just the cross-section: within a firm, payout should follow slack over time, and investment should too. That is a harder and better test, and it needs panel data rather than a cross-section.
- Nothing here is causally identified. What the model buys is not proof — it is a **specification**: it tells you which variable was missing and why.

Which is the honest reason to build the theory: not to win the argument, but to know what to put on the right-hand side.

The frictions ledger — Week 43

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Seven rows — and this one is different in kind. Every other friction is a **state of the world**. This one is a **position**, and it moves.

Seven weeks, read as one map

- Three families: **prices are wrong** (36, 38) · **somebody else decides** (39, 41, 42) · **the future constrains today** (37, 43).
- And one object kept reappearing under different names. **Week 38's net worth. Week 42's bank capital. This week's manager's stake.** Each time: somebody's own money, at risk, deciding what the firm may do next.
- The difference today is that we did not **assume** it mattered. We watched it accumulate, and everything else followed.

The three families are my grouping, not the literature's. But the point underneath them is: not seven topics — **one question**, asked seven times.

What we still cannot explain

- Week 36 opened with Lemmon et al. (2008): most variation in leverage is an **unexplained firm fixed effect**. Seven weeks later that is still true.
- These models are **qualitative**. They tell you the sign and the shape — payout waits, investment tracks slack, the wedge moves — and almost never the magnitude.
- And they are hard to reject. A model with an unobserved state variable explains a great deal, which is a strength and a warning at the same time.

The direction is settled. The **magnitude** is not — which is why this course made you reproduce a table rather than memorise one.

Reading, and what happens next

- **Reading:** DeMarzo and Sannikov (2006) — the introduction and the implementation section, **for the picture, not the equations**. The paper is technical and you are not being asked to solve it.
- **Evidence:** DeAngelo et al. (2006), read **like a referee** with the age-and-size objection in hand.
- **This week:** **individual reflections** open — graded per student, not per team, unlike the project itself. The question is where the AI was **wrong** and how you caught it — a reflection with no error in it is not a reflection.
- **The exam:** closed-book, calculator, calculation-heavy — every week's two-state model, friction identification, and reading a real table.

A dividend is what a firm pays when it has finally accumulated enough to be trusted with the rest.

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